

NTT Examples

Cyclic and Negacyclic Convolution in $\mathbb{F}_{17}$

SEC 595: Encrypted Computing

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The Setting

Everything below happens in one small field, chosen so that every step can be checked by hand.

Quantity	Value
Field	$\mathbb{F}_{17}$, so $q = 17$
Two-adicity	$q - 1 = 16 = 2^4$, so $2^4 \mid q - 1$
Dimension	$N = 4$
Primitive N -th root	$\omega = 13$, since $13^4 \equiv 1$ and $13^2 \equiv 16 \equiv -1$
Primitive $2N$ -th root	$\psi = 9$, since $9^2 = 81 \equiv 13 = \omega$
Inverses	$\omega^{-1} = 4$, $\psi^{-1} = 2$, $N^{-1} = 4^{-1} \equiv 13$

The two input polynomials throughout are

$$a(X) = 1 + 2X + 3X^2 + 4X^3, \quad b(X) = 4 + 3X + 2X^2 + X^3.$$

The ordinary product over $\mathbb{Z}$

Before any reduction, convolve the coefficient vectors $(1, 2, 3, 4)$ and $(4, 3, 2, 1)$. The result is the seven-term sequence

$$(d_0, d_1, d_2, d_3, d_4, d_5, d_6) = (4, 11, 20, 30, 20, 11, 4),$$

for instance $d_3 = 1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3 + 4 \cdot 4 = 30$.

The terms d_4, d_5, d_6 carry degrees 4, 5, 6 and have nowhere to go in a length-4 answer. **What happens to them is the only difference between the two parts below.**

Part 1. Standard NTT: Cyclic Convolution modulo $X^4 - 1$

The wrapping rule

Modulo $X^4 - 1$ we have $X^4 \equiv 1$, so a wrapped term returns with a **positive** sign:

$$c_k = d_k + d_{k+4}.$$

Directly:

$$\begin{aligned} c_0 &= 4 + 20 = 24 \equiv 7, & c_1 &= 11 + 11 = 22 \equiv 5, \\ c_2 &= 20 + 4 = 24 \equiv 7, & c_3 &= 30 \equiv 13, \end{aligned}$$

so $ab = (7, 5, 7, 13)$ in $\mathbb{F}_{17}[X]/(X^4 - 1)$.

The same answer via the transform

The NTT evaluates a polynomial at the powers of ω , which for $\omega = 13$ are

$$(\omega^0, \omega^1, \omega^2, \omega^3) = (1, 13, 16, 4).$$

Step 1 — Transform. Evaluate both polynomials at those four points, $\hat{a}_k = a(\omega^k)$:

$$\text{NTT}(a) = (10, 6, 15, 7), \quad \text{NTT}(b) = (10, 11, 2, 10).$$

For instance $a(\omega) = 1 + 2(13) + 3(16) + 4(4) = 1 + 26 + 48 + 16 = 91 \equiv 6 \pmod{17}$.

Step 2 — Pointwise multiplication. Multiply the two vectors entrywise:

$$(10 \cdot 10, 6 \cdot 11, 15 \cdot 2, 7 \cdot 10) = (100, 66, 30, 70) \equiv (15, 15, 13, 2) \pmod{17}.$$

Step 3 — Inverse transform. Apply NTT^{-1} , which is the same evaluation with $\omega^{-1} = 4$ in place of ω , scaled by $N^{-1} = 13$:

$$\text{NTT}^{-1}(15, 15, 13, 2) = (7, 5, 7, 13).$$

This agrees with the direct computation. Three transforms of $O(N \log N)$ each replace the $O(N^2)$ convolution.

Part 2. Negacyclic NTT: Convolution modulo $X^4 + 1$

Lattice-based cryptography — ML-KEM, ML-DSA, and every FHE scheme in this course — works modulo $X^N + 1$, not $X^N - 1$. One sign changes, and the transform has to be rebuilt.

The wrapping rule

Modulo $X^4 + 1$ we have $X^4 \equiv -1$, so a wrapped term returns with a **negative** sign:

$$c_k = d_k - d_{k+4}.$$

Directly:

$$\begin{aligned} c_0 &= 4 - 20 = -16 \equiv 1, & c_1 &= 11 - 11 = 0, \\ c_2 &= 20 - 4 = 16 \equiv -1, & c_3 &= 30 \equiv 13 \equiv -4, \end{aligned}$$

so $ab = (1, 0, 16, 13)$ in $\mathbb{F}_{17}[X]/(X^4 + 1)$, which in centered representatives is $(1, 0, -1, -4)$.

Why the plain transform will not do

The NTT of Part 1 computes $c_k = d_k + d_{k+4}$, and no choice of evaluation points changes that: ω has order 4, so $\omega^4 = 1$ is exactly the relation $X^4 = 1$ being imposed. To impose $X^4 = -1$ instead, we need a root of order $2N = 8$, namely $\psi = 9$ with $\psi^2 = \omega$. Its powers are

$$(\psi^0, \psi^1, \psi^2, \psi^3) = (1, 9, 13, 15), \quad (\psi^0, \psi^{-1}, \psi^{-2}, \psi^{-3}) = (1, 2, 4, 8).$$

The five steps

Step 1 — Twist. Scale the i -th coefficient by ψ^i , giving $\tilde{a}_i = \psi^i a_i$ and $\tilde{b}_i = \psi^i b_i$:

$$\tilde{a} = (1 \cdot 1, 9 \cdot 2, 13 \cdot 3, 15 \cdot 4) = (1, 18, 39, 60) \equiv (1, 1, 5, 9),$$

$$\tilde{b} = (1 \cdot 4, 9 \cdot 3, 13 \cdot 2, 15 \cdot 1) = (4, 27, 26, 15) \equiv (4, 10, 9, 15).$$

Step 2 — Transform. Run the *same* length-4 NTT as Part 1, at the same points $(1, 13, 16, 4)$, on the twisted vectors:

$$\text{NTT}(\tilde{a}) = (16, 11, 13, 15), \quad \text{NTT}(\tilde{b}) = (4, 15, 5, 9).$$

Step 3 — Pointwise multiplication.

$$(16 \cdot 4, 11 \cdot 15, 13 \cdot 5, 15 \cdot 9) = (64, 165, 65, 135) \equiv (13, 12, 14, 16) \pmod{17}.$$

Step 4 — Inverse transform. Again the standard NTT^{-1} , with $\omega^{-1} = 4$ and $N^{-1} = 13$:

$$\text{NTT}^{-1}(13, 12, 14, 16) = (1, 0, 4, 8).$$

Step 5 — Untwist. Scale the i -th coefficient by ψ^{-i} , using the powers $(1, 2, 4, 8)$:

$$(1 \cdot 1, 0 \cdot 2, 4 \cdot 4, 8 \cdot 8) = (1, 0, 16, 64) \equiv (1, 0, 16, 13) \pmod{17}.$$

This is the negacyclic product computed directly above, in centered form $(1, 0, -1, -4)$.

The Two Side by Side

	Cyclic , $X^4 - 1$	Negacyclic , $X^4 + 1$
Wrapping rule	$c_k = d_k + d_{k+4}$	$c_k = d_k - d_{k+4}$
Root needed	ω of order N	ψ of order $2N$, $\psi^2 = \omega$
Condition on q	$N \mid q - 1$	$2N \mid q - 1$
Extra work	none	twist before, untwist after
Result	$(7, 5, 7, 13)$	$(1, 0, 16, 13) = (1, 0, -1, -4)$
Used by	signal processing	ML-KEM, ML-DSA, BFV, BGV, CKKS

The twist and untwist cost $2N$ extra multiplications, which is negligible beside the $O(N \log N)$ of the transform itself. What is *not* negligible is the condition in the third row: asking for a root of order $2N$ rather than N is what forces $q \equiv 1 \pmod{2N}$, and that single congruence is why ML-KEM and ML-DSA choose the moduli they do.

Exercises

1. Verify $\text{NTT}(a) = (10, 6, 15, 7)$ by evaluating a at each of $1, 13, 16, 4$.
2. Confirm that $\psi = 9$ has order exactly 8 in $\mathbb{F}_{17}^\times$ by listing its first eight powers.
3. Compute $a(X)^2$ modulo $X^4 + 1$ over $\mathbb{F}_{17}$, once directly and once by the five-step route.

4. Show that $\psi^4 \equiv -1 \pmod{17}$, and explain why this is the identity that makes the twist impose $X^4 = -1$.
5. The prime $q = 13$ has $q - 1 = 12$, so $4 \mid q - 1$ but $8 \nmid q - 1$. Which of the two transforms above can be run over $\mathbb{F}_{13}$ at $N = 4$, and which cannot?